Spectrum Analysis of Sub-1 GHz, Mid-band & mmWave by Plotting Propagation

1. Aim

To analyze the propagation loss characteristics of different frequency bands (Sub-1 GHz, Midband, and mmWave) in wireless communication systems and compare their performance over varying distances using MATLAB.

2. Software Used

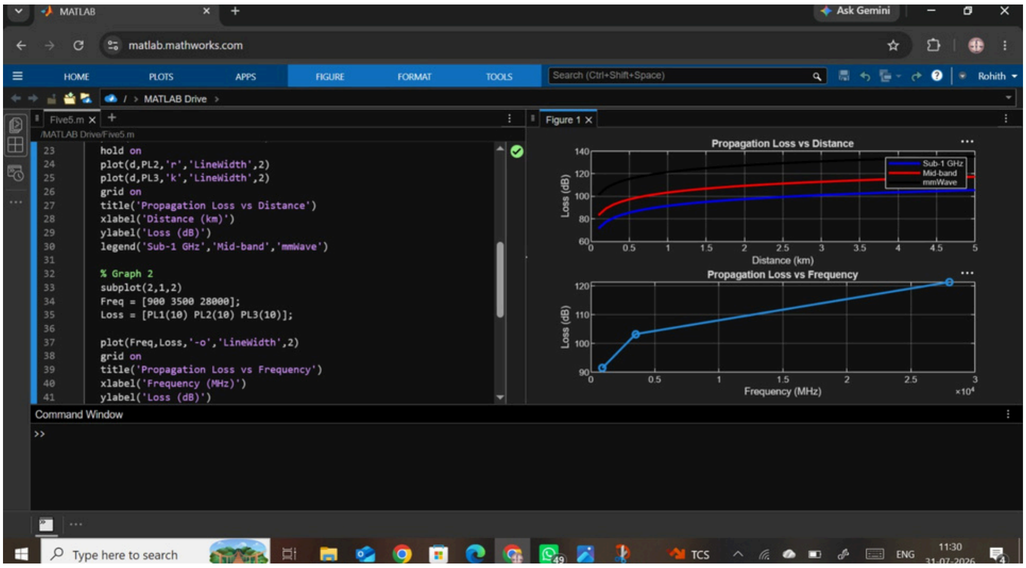
MATLAB

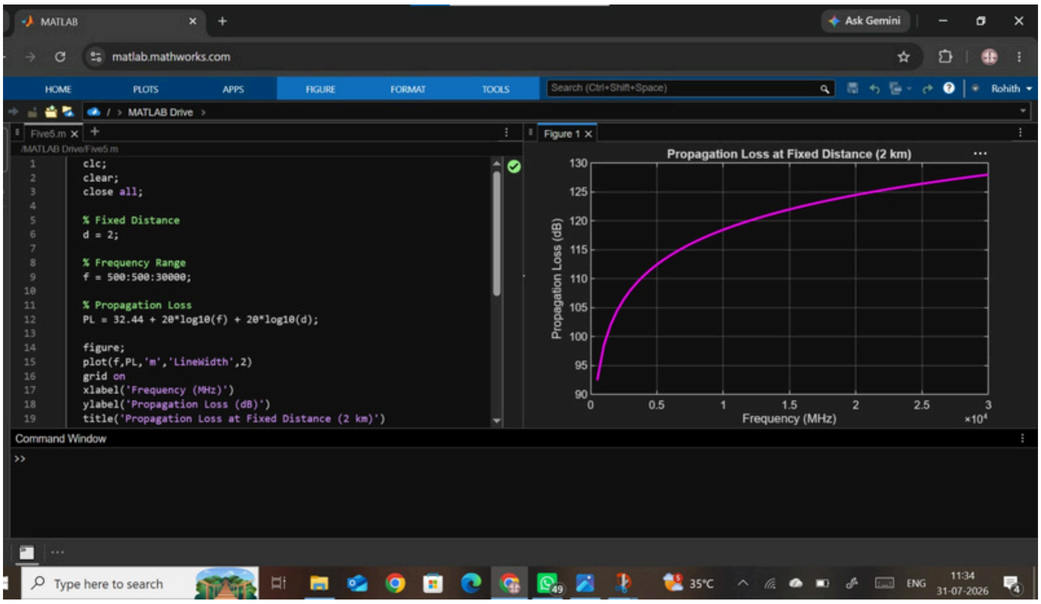
3.Code

```
clc;
clear;
close all;

%% Objective 1

% Distance (km)
d = 0.1:0.1:5;

% Frequencies (MHz)
f = [900 3500 28000];

% Free Space Path Loss
PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);

figure;

% Graph 1
subplot(2,1,1)
plot(d,PL1,'b','LineWidth',2)
hold on
plot(d,PL2,'r','LineWidth',2)
plot(d,PL3,'k','LineWidth',2)
grid on
title('Propagation Loss vs Distance')
```

```matlab
% Distance (km)
d = 0.1:0.1:5;

% Frequencies (MHz)
f = [900 3500 28000];

% Free Space Path Loss
PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);

figure;

% Graph 1
subplot(2,1,1)
plot(d,PL1,'b','LineWidth',2)
hold on
plot(d,PL2,'r','LineWidth',2)
plot(d,PL3,'k','LineWidth',2)
grid on
title('Propagation Loss vs Distance')
xlabel('Distance (km)')
ylabel('Loss (dB)')
legend('Sub-1 GHz','Mid-band','mmWave')

% Graph 2
subplot(2,1,2)
Freq = [900 3500 28000];
Loss = [PL1(10) PL2(10) PL3(10)];

plot(Freq,Loss,'-o','LineWidth',2)
grid on
title('Propagation Loss vs Frequency')
xlabel('Frequency (MHz)')
ylabel('Loss (dB)')
```

6. Output Graphs





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7. Result

The propagation loss of Sub-1 GHz (900 MHz), Mid-band (3.5 GHz), and mmWave (28 GHz) was successfully analyzed using MATLAB. The results show that propagation loss increases with both communication distance and operating frequency. Among the three bands, mmWave exhibits the highest propagation loss, Mid-band has moderate loss, and Sub-1 GHz provides the lowest propagation loss and the best coverage. This demonstrates the trade-off between coverage and capacity in different 5G spectrum bands.
